

Collin Christy | Curriculum Vitae

Steward Observatory, University of Arizona

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Research Interests

I use radio, millimeter, and X-ray observations to study energetic extragalactic transients, with a focus on tidal disruption events, gamma-ray bursts, and unusual supernovae. My research probes the physics of relativistic jets and other outflows powered by accretion onto compact objects.

Education

University of Arizona **Tucson, AZ**

Ph.D. in Astronomy and Astrophysics

Thesis: *Characterizing the Diversity and Energetics of Relativistic Outflows in Extragalactic Transients*

Advisor: Kate Alexander, Ph.D.

University of Arizona **Tucson, AZ**

M.A. in Astronomy and Astrophysics

2024

The Ohio State University **Columbus, OH**

B.S. in Physics and Astronomy & Astrophysics, Minor in Mathematics, Summa Cum Laude 2021

Related Employment

University of Arizona **Supervisor: Kate Alexander, Ph.D.**

Graduate Research Assistant

2022–Present

All-Sky Automated Survey for SuperNovae

Data Analyst

2021–2022

The Ohio State University

Research Assistant

Supervisor: Krzysztof Stanek, Ph.D.

2020–2021

Telescope Time Allocations (as PI)

VLA: 6 Proposals – 67 hr

GMRT: 3 Proposals – 30 hr

ALMA: 1 Proposal – 3 hr

MMT: 4 Proposals – 18 nights

LBT: 4 Proposals – 2 nights

Bok 90": 2 Proposals – 4 nights

Conference Contributions and Talks

2024: *Radio Evolution of the Tidal Disruption Event ASASSN-19bt: Evidence for a Highly Off-Axis Relativistic Jet*, Kavli Institute for Theoretical Physics (Talk)

2022: *Variable Star Classification Using Citizen Science and the All-Sky Automated Survey for SuperNovae (ASAS-SN)*, Spoken-WERRD (Talk)

2021: *Citizen ASAS-SN: Citizen Science with The All-Sky Automated Survey for SuperNovae (ASAS-SN)*, AAS 237th Meeting (iPoster)

2020: *ASAS-SN Variable Star Classification using Machine Learning*, Big Data Analytics in Physics, The Ohio State University (Poster)

2020: *The All-Sky Automated Survey for SuperNovae (ASAS-SN) Citizen Science Project*, Big Data Analytics in Physics, Summer Undergraduate Research Program Symposium (SURP), The Ohio State University (Talk)

Peer Review

Reviewer: Giant Meterwave Radio Telescope

Referee: Astrophysical Journal

Referee: Astrophysical Journal Letters

2024 –

2024 –

2024 –

Awards

2024: NSF Graduate Research Fellowship Honorable Mention

2018: The Ohio State University Helen Cowan Book Award

Technical Skills

Programming: Python, C++, R, LaTeX, HTML, Mathematica

Data Reduction: CASA, Miriad, AIPS

Systems: Linux, macOS, Windows

Observing Experience

Radio: VLA, GMRT, ATA, ALMA, ATCA, MeerKAT, NOEMA, SMA

Optical: Bok, MMT, LBT, ASAS-SN

Publications

Author summary: 24 total publications with >280 citations, including 6 first-author papers. A full list is available on the NASA ADS.

First Author Publications

- Christy, Collin T., Laskar, Tanmoy, Alexander, Kate D. et al. (2026), *First Detection of Faraday Rotation in a Gamma-Ray Burst Afterglow: Low Polarization and High Rotation Measure in GRB 260310A Reveal Jet Magnetic Structure and Environment*, arXiv e-prints (arXiv:2604.27480).
- Christy, Collin T., Alexander, Kate D., Laskar, Tanmoy et al. (2026), *Dichotomy in Long-Lived Radio Emission from Tidal Disruption Events AT 2020zso and AT 2021sdu: Multi-Component Outflows vs. Host Contamination*, ApJ, 1001
- Christy, Collin T., Alexander, Kate D., Margutti, Raffaella et al. (2024), *The Peculiar Radio Evolution of the Tidal Disruption Event ASASSN-19bt*, ApJ, 974.
- Christy, C. T., Jayasinghe, T., Stanek, K. Z. et al. (2023), *The ASAS-SN catalogue of variable stars X: discovery of 116 000 new variable stars using G-band photometry*, MNRAS, 519.
- Christy, C. T., Jayasinghe, T., Stanek, K. Z. et al. (2022), *Citizen ASAS-SN Data Release. I. Variable Star Classification Using Citizen Science*, PASP, 134.
- Christy, C. T., Jayasinghe, T., Stanek, K. Z. et al. (2021), *Citizen ASAS-SN: Citizen Science with The All-Sky Automated Survey for SuperNovae (ASAS-SN)*, Research Notes of the American Astronomical Society, 5, 38.

Publications as Co-Author.....

- Franz, Noah et al. (2025), *Optimizing Kilonova Searches: A Case Study of the Type IIb SN 2025ulz in the Localization Volume of the Low-Significance Gravitational Wave Event S250818k*, arXiv e-prints (arXiv:2510.17104).
- Sfaradi, Itai et al. (2025), *The First Radio-bright Off-nuclear Tidal Disruption Event AT 2024tvd Reveals the Fastest-evolving Double-peaked Radio Emission*, ApJL, 992, L18.
- Burn, Matthew et al. (2025), *The 6 year radio lightcurve of the tidal disruption event AT2019azh*, arXiv e-prints (arXiv:2509.17525).
- Franz, Noah et al. (2025), *The Open multiwavelength Transient Event Repository (OTTER): Infrastructure Release and Tidal Disruption Event Catalog*, arXiv e-prints (arXiv:2509.05405).
- Nayana, A. J. et al. (2025), *The Most Luminous Known Fast Blue Optical Transient AT 2024wpp: Unprecedented Evolution and Properties in the X-rays and Radio*, arXiv e-prints (arXiv:2509.00952).
- Bostroem, K. Azalee et al. (2025), *Late-time Hubble Space Telescope Ultraviolet Spectra of SN 2023ixf and SN 2024ggi Show Ongoing Interaction with Circumstellar Material*, arXiv e-prints (arXiv:2508.11756).
- Sfaradi, Itai et al. (2025), *The First Radio-Bright Off-Nuclear TDE 2024tvd Reveals the Fastest-Evolving Double-Peaked Radio Emission*, arXiv e-prints (arXiv:2508.03807).
- Kwok, Lindsey A. et al. (2025), *JWST and Ground-based Observations of the Type Iax Supernovae SN 2024pxl and SN 2024vjm: Evidence for Weak Deflagration Explosions*, ApJL, 989 (arXiv:2505.02944).
- Cendes, Yvette et al. (2025), *Continued Rapid Radio Brightening of the Tidal Disruption Event AT2018hyz*, arXiv e-prints (arXiv:2507.08998).
- Yao, Yuhan et al. (2025), *Optically Overluminous Tidal Disruption Events: Outflow Properties and Implications for Extremely Relativistic Disruptions*, arXiv e-prints (arXiv:2507.06453).
- Alexander, Kate D. et al. (2025), *The Multi-Wavelength Context of Delayed Radio Emission in TDEs: Evidence for Accretion-Driven Outflows*, arXiv e-prints (arXiv:2506.12729).
- Singh, Mridweeka et al. (2025), *Photometry and Spectroscopy of SN 2024pxl: A Luminosity Link Among Type Iax Supernovae*, arXiv e-prints (arXiv:2505.02943).
- Hajela, A. et al. (2025), *Eight Years of Light from ASASSN-15oi: Toward Understanding the Late-time Evolution of TDEs*, ApJ, 983 (arXiv:2407.19019).
- Goodwin, A. J. et al. (2025), *A Second Radio Flare from the Tidal Disruption Event AT2020vwl: A Delayed Outflow Ejection?*, ApJ, 981 (arXiv:2410.18665).
- Andrews, Jennifer E. et al. (2025), *Asymmetries and Circumstellar Interaction in the Type II SN 2024bch*, ApJ, 980 (arXiv:2411.02497).
- Kotrach, O. et al. (2024), *Citizen ASAS-SN Data Release II: Variable Star Classification Using Citizen Science*, arXiv e-prints (arXiv:2408.07859).
- Phillips, Anya et al. (2024), *Seven classes of rotational variables from a study of 50 000 spotted stars with ASAS-SN, Gaia, and APOGEE*, MNRAS, 527 (arXiv:2305.09715).
- Laskar, Tanmoy et al. (2023), *The Radio to GeV Afterglow of GRB 221009A*, ApJL, 946 (arXiv:2302.04388).